



DPUFader

Fading signals in and/or out.

1. Inputs:

<i>In</i>	Continuous input
<i>Fade_In</i>	Continuous fade input which is multiplied with the current value of the fade function (<i>FadeInFunc</i> , <i>FadeOutFunc</i>). It is added to the value of Input <i>In</i> .
<i>FadeInTime</i>	Duration of fading process in milliseconds
<i>FadeInFunc</i>	Signal curve of fading process
<i>FadeHoldTime</i>	Hold time of fade in milliseconds
<i>FadeOutTime</i>	Duration of fade out in milliseconds
<i>FadeOutFunc</i>	Signal curve of fade out
<i>Start</i>	If value > 0 in/-out fading starts.

2. Outputs:

<i>Out</i>	Output value
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3. Functional principle:

The current value of *Fade_In* is multiplied with the current value of the fade function and added to the input *In*. The result is available at output *Out*.

The fade function for fading in/out of *Fade_In* can be set using the parameters *FadeInFunc* and *FadeOutFunc* and is computed as following:

Fade in:

$$\left(\left(\frac{FadeInTime_{End} - FadeInTime_{Start}}{FadeInTime} \right)^{FadeInFunc} \right) \cdot Fade_In$$

Fade out:

$$\left(\left(1 - \frac{FadeOutTime_{End} - FadeOutTime_{Start}}{FadeOutTime} \right)^{FadeOutFunc} \right) \cdot Fade_In$$

The fade-in process is triggered by the positive edge of input signal *Start*.



SILAB DRIVING SIMULATION

version Version 7.1 documentation

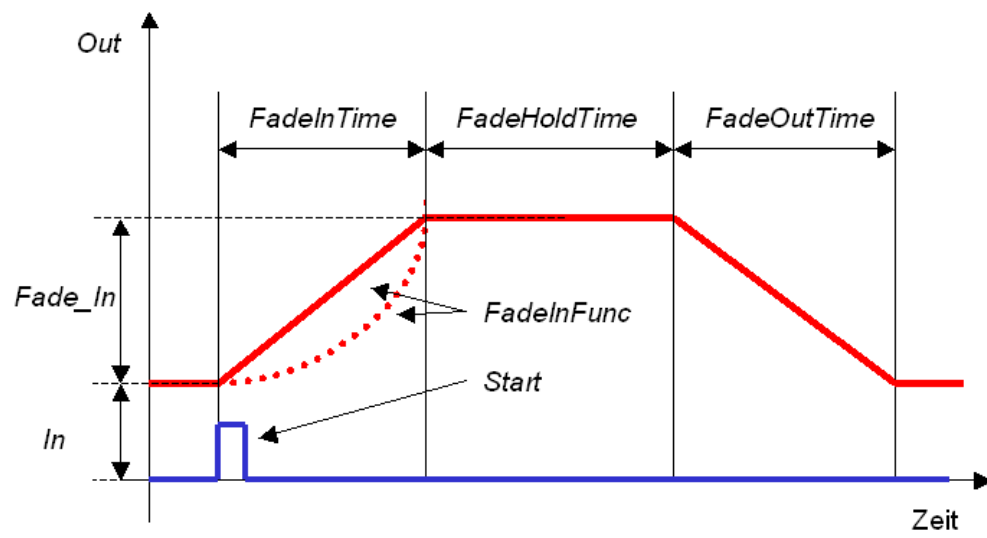


Figure: Process at constant In and Fade_In.